

MoMo SMS ERD Documentation

The Mobile Money SMS Data processing system was designed to process MoMo SMS data in XML format. The system has an Entity Relationship Diagram that defines entities like USERS, SMS_MESSAGES, TRANSACTIONS, TRANSACTION_CATEGORIES, SYSTEM_LOG, and TRANSACTION_PARTICIPANTS.

The ERD's purpose was to format, clean, and categorize the MoMo SMS data so that we store it in a relational database. It defines entities, relationships between entities, cardinality, and relevant attributes for each entity, including primary and foreign keys.

In the ERD, the following assumptions were made:

- System logs are generated automatically.
- Each user's MoMo is an account in itself, omitting the need for an extra account entity.
- The currency is uniform across all transactions.
- Each transaction belongs to one category.

Entity Descriptions

USERS: The MoMo account owners.

SMS_BACKUPS: The XML data structure file from which data was extracted.

SMS-MESSAGES: Each message from the XML data structure.

TRANSACTION_PARTICIPANTS: Entities involved in the various transactions.

CONTACTS: People the MoMo account owner interacts with.

SYSTEM_LOGS: Machine-generated records to track data processing

TRANSACTIONS: Main transaction records

TRANSACTIONS_CATEGORIES: Categories like withdrawal, payment, receiving money, etc

A many-to-many relationship exists between the TRANSACTIONS and CONTACTS entities, since one transaction can involve many contacts and one contact can appear in many transactions. To resolve this, the

TRANSACTIONS_PARTICIPANTS junction table was introduced. Each record in it links one transaction to one contact and specifies the role that the contact played in that transaction. This turns the many-to-many relationship into two one-to-many relationships. This design eliminates redundancy from the database.

In the future, the ERD could evolve to support multiple accounts and different currencies for each user.